

Rocky's Algorithmic Adventures



A Fun Guide to Coding and Logic!

Page 2: The Teacher's Vocabulary Chest

1. **Algorithm**  - step-by-step instructions
2. **Sequencing**  - putting steps in the correct order
3. **Precision**  - being exact and clear
4. **Decomposition**  - breaking problems down
5. **Debugging**  - finding and fixing mistakes.

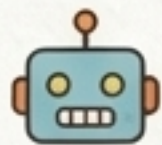


Page 3: Pedagogical Guide - How to Teach Logic

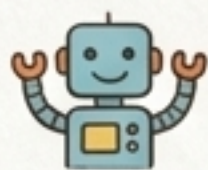
- 💡 1. Build an Error-Positive Culture (bugs are awesome!)
- 🧑 2. Start Kinesthetic (move bodies first)
- 💡 3. Ask Guiding Questions (e.g., Where is Kojo looking?)
- 👥 4. Encourage Peer Collaboration (work in pairs)



Page 4: The Silly PB&J Robot



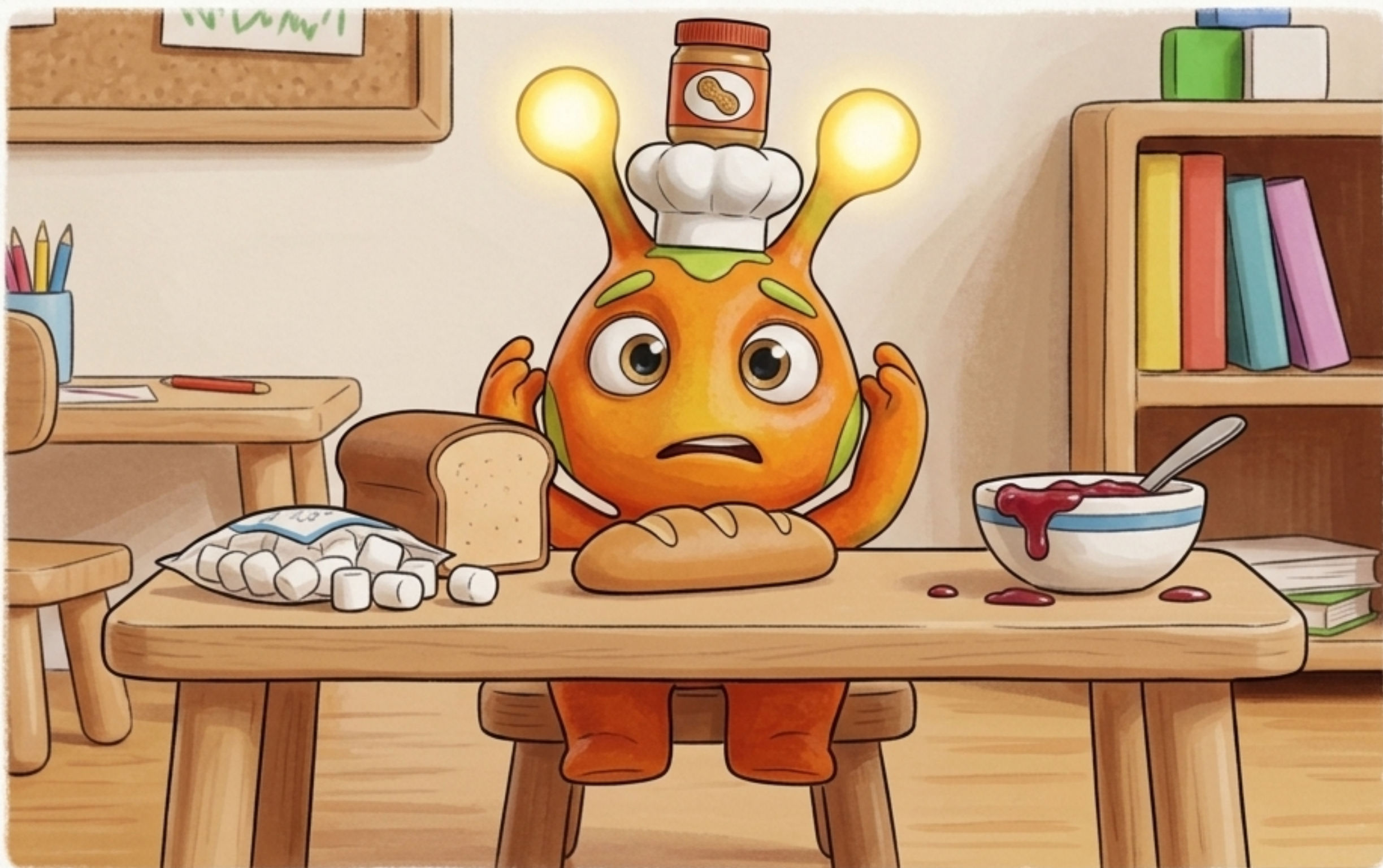
Concept: Robots need precise instructions.



Activity: The teacher acts as a Robot. Children give step-by-step instructions to make a peanut butter sandwich.



The Bug: If directions are too general (e.g. “Put jar on bread”), the robot does silly things! Children must debug the recipe.



Page 5: The Human Robot Game

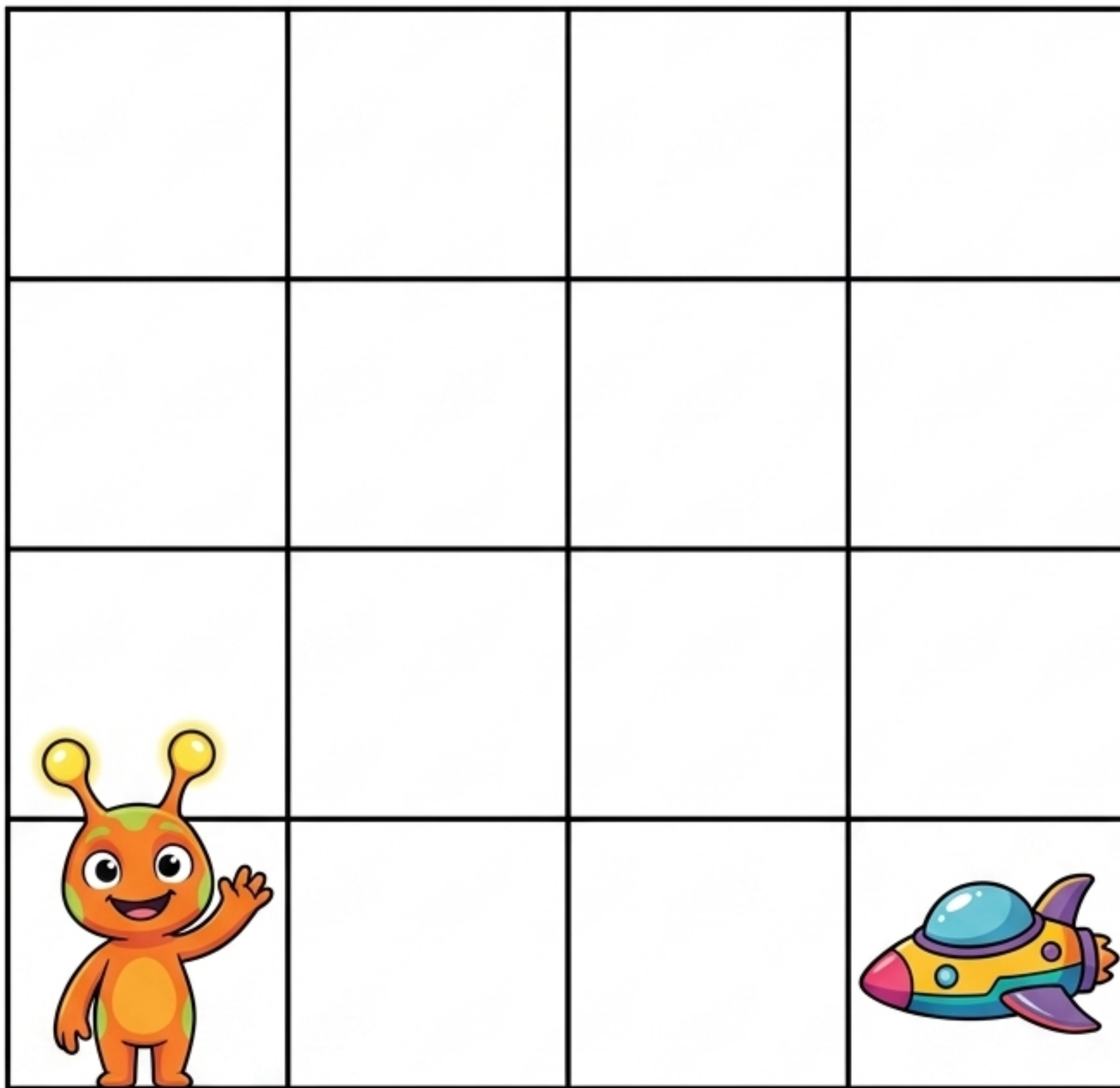
Game Rules:

- 1. Setup:** Create a grid mat on the floor.
- 2. Roles:** One child is the Programmer, one is the Robot.

- 3. Goal:** The Programmer uses coding cards (Forward, Turn Right, Turn Left) to guide the Robot to the target.
- 4. Debugging:** Celebrate mistakes! Find the bug and correct it together.



Level 1: Basic Path Planning



Help Rocky find his way to the spaceship!
 Draw a path on the grid. He can only move UP and RIGHT.
 Plan his journey!

NAME:

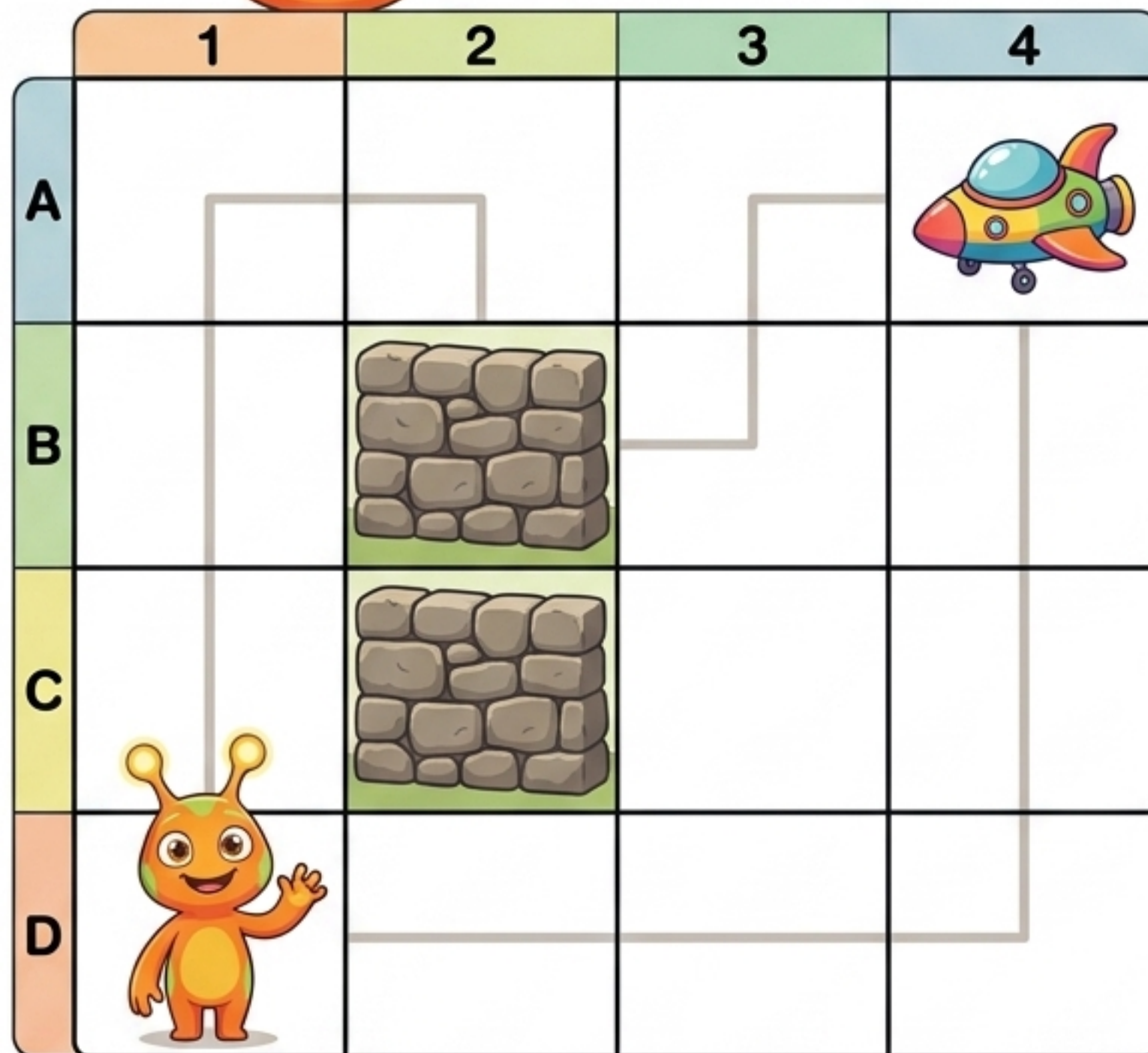


DATE:

Level 2: Bypassing Obstacles



Help me find my spaceship!



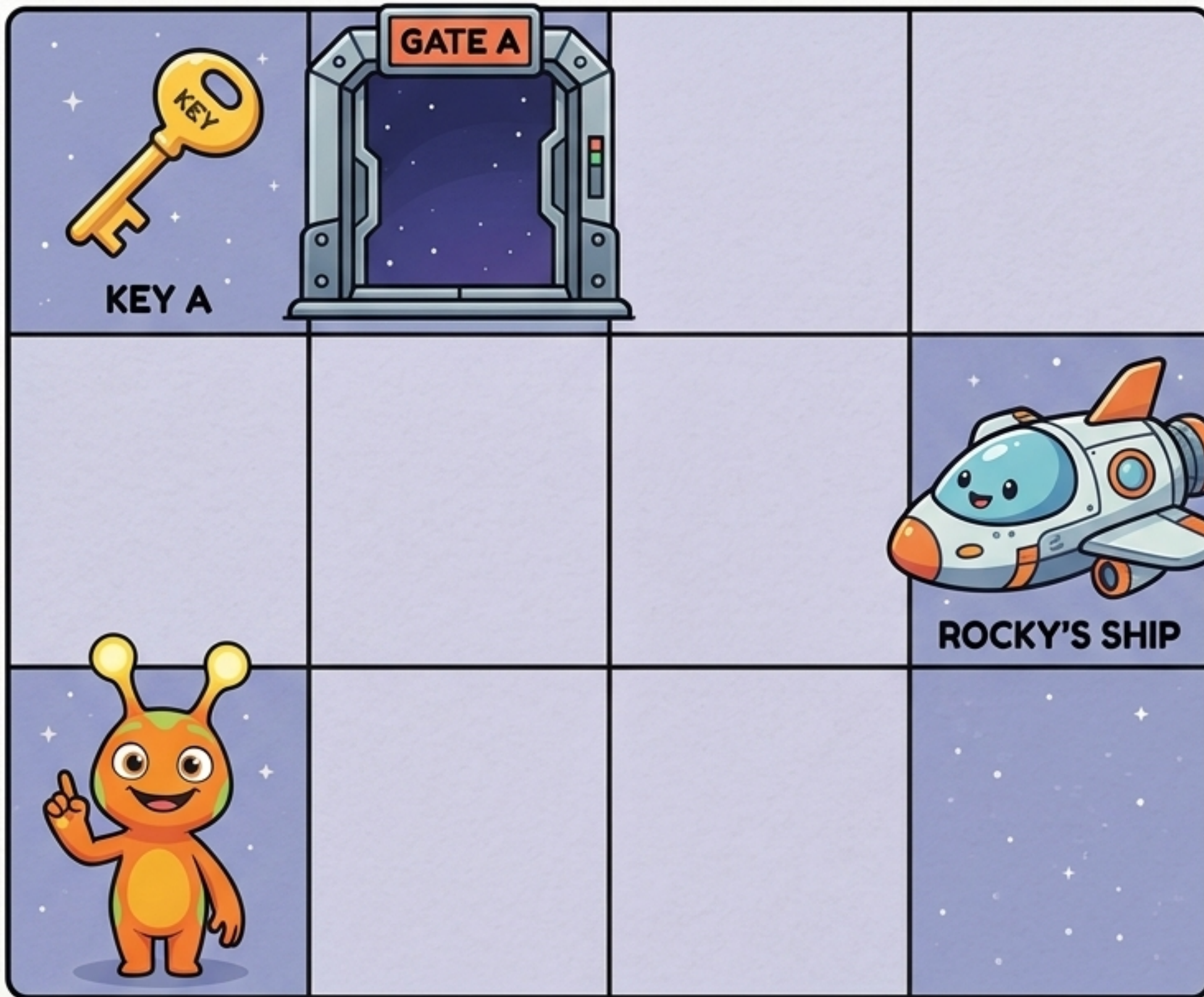
Draw your path!



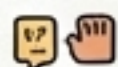
Problem Solved?

☐☐☐

Level 3: Checkpoint Sequencing



INSTRUCTIONS:



1. Help Rocky get to his ship! Draw arrows on the grid to show his path.



2. What order must Rocky visit the items?

3. Draw the path order in the circles below:

1

2

3

4



Key A



Gate A



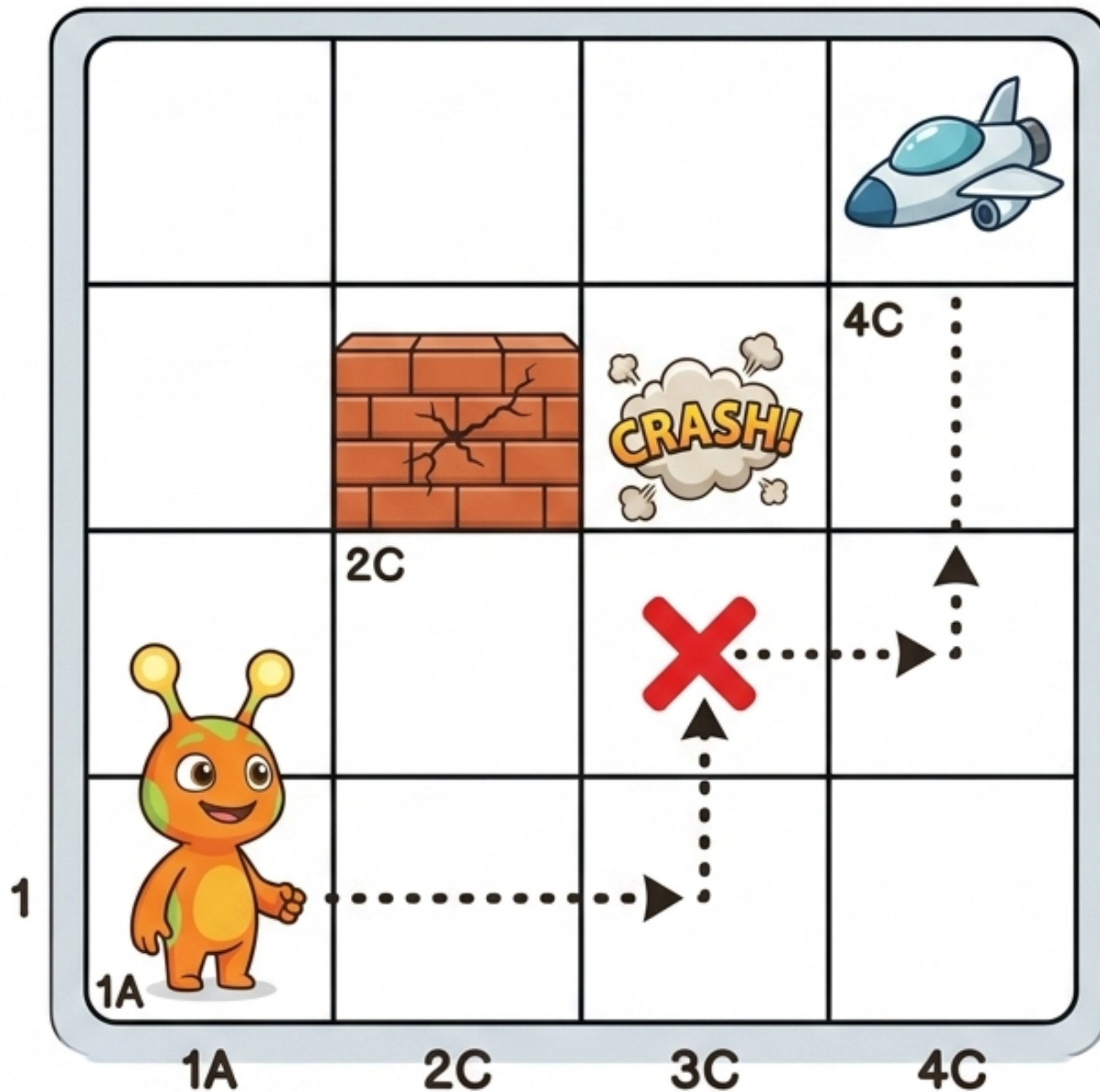
Ship



Rocky (Start)

Level 4: Hit the Wall

DEBUGGING PUZZLE



Buggy Algorithm

1. Forward

2. Forward

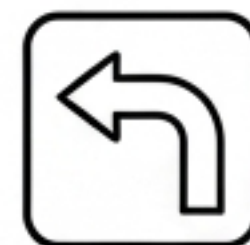
3. Turn Right

4. Forward

5. Forward

Can you spot and fix the bug in this sequence?

How to Fix It



Left Turn



U-Turn



Level 5: The Missed Checkpoint



Cols Rows	A	B	C	D
1				
2		B2 		D2 
3		C1		D1
4	A1 			

Dashed path with arrows: A1 → C1 → B2 → D1 → D2

COMMANDS



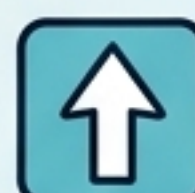
Turn Right



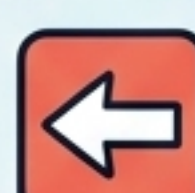
& Text



Forward



& Text



& Text



& Text



CHECKPOINT STATUS



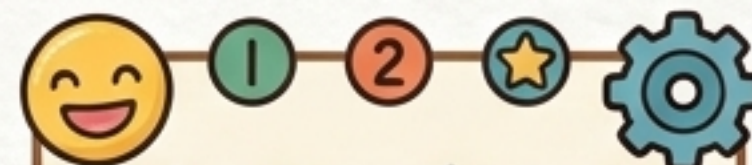
**MISS
CHECKPOINT!**

Can you fix the code to collect Key A?



Level 6: Path Optimization

	1	2	3	4
1				
2				
3				
4				



Instructions:
Help Rocky find the shortest route to Kojo!
Write the step-by-step path below.

Answering

Step 1: _ _ _ _ _

Step 2: _ _ _ _ _

Step 3: _ _ _ _ _

Step 4: _ _ _ _ _

Step 5: _ _ _ _ _





FORWARD

BACKWARD

TURN LEFT

TURN RIGHT

PLANET LOGIX

DIRECTION COMMANDS

▶ PLAY



LEARN & DANCE!

Educational Vector Art for Kids

THE ALGORITHM SONG



SING ALONG & LEARN WITH ROCKY!

TURN LEFT, TURN RIGHT

Educational Song & Activity



MOVE and GROOVE: THE EDUCATIONAL DANCE SONG!



EDUCATE, EXERCISE, ENTERTAIN – PULL IT ALL TOGETHER!

A Logix Learning Adventure

Let's Move Together!

Featuring Rocky & Kojo
An Educational Dance-Along Song!

